

Objective

Metrics

Reference

Simplified 'No-Judge' evaluation for the College Chatbot

Overview — Objective Metrics Only

Following user feedback, the qualitative LLM Judge has been removed. Evaluation is now 100% objective, focusing on semantic truth and routing accuracy.

Metric	Sheet	Weight	Goal
Time Taken (s)	Production	—	Fast server processing
Context	Both	—	Visible facts used for answer
BERTScore F1 %	Evaluation	50 %	Semantic truth vs context
Source Score	Evaluation	25 %	Correct system routing
Link Validity	Evaluation	25 %	Live resource check
Accuracy %	Evaluation	Total	Weighted objective score

SECTION 1

Performance & Transparency

Time Taken (s)

PROD

WHAT IT MEASURES

Server processing time.

FORMULA

T_end - T_start

HOW IT WORKS

Calculated on server-side.

SCORE	INTERPRETATION
0-2s	Great
>5s	Slow

■ Background logging ensures zero impact on user.

Context

BOTH

WHAT IT MEASURES

Raw data retrieved to answer the question.

FORMULA

KB Facts | SQL Rows | PDF Snippets

HOW IT WORKS

Passed from backend to log and test suite.

SCORE	INTERPRETATION
Present	Transparent
Empty	Review

This is the primary transparency tool.

SECTION 2

Objective Quality Metrics

BERTScore F1 %

EVAL

WHAT IT MEASURES

Semantic similarity between answer and context.

FORMULA

F1 (BERT-style) mapping

HOW IT WORKS

Uses all-MiniLM-L6-v2 sentence embeddings.

SCORE	INTERPRETATION
80-100%	Accurate
<50%	Suspect

■ Weighted at 50% of Final Accuracy.

Link Validity

EVAL

WHAT IT MEASURES

Whether links in the answer are live.

FORMULA

X/Y Valid

HOW IT WORKS

Pings URLs via HTTP HEAD/GET.

SCORE	INTERPRETATION
All Valid	Good
Partial	Stale

■ Weighted at 25% of Final Accuracy.

SECTION 3

Final Accuracy

Accuracy %

EVAL

WHAT IT MEASURES

Total balanced quality score.

FORMULA

$BERT(0.5) + Source(0.25) + Links(0.25)$

HOW IT WORKS

Weighted average of all objective checks.

SCORE	INTERPRETATION
85-100%	Reliable
<65%	Fix Needed

■ Strictly objective; no LLM judge bias.